

OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-39 — D-Mannose Complex Capsules**, analysed EU-wide (Pan-EU). Prepared for review.

1. Data sources

- **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
- **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
- **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is $P(0)=e^{-\lambda}$: ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).

Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	$3 \leq \text{reach} < 30$ days, still selling (units > 0), throttled (price $\geq +8\%$ and/or ad cut $\geq 50\%$)	Miss (voluntary)
4	Demand gap (EU)	Units = 0 on a day enclosed by sales, $\lambda \geq 3/\text{day}$ (incl. throttles that hit 0)	Lost (involuntary)

Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = $\max(\lambda - \text{actual units}, 0)$** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:

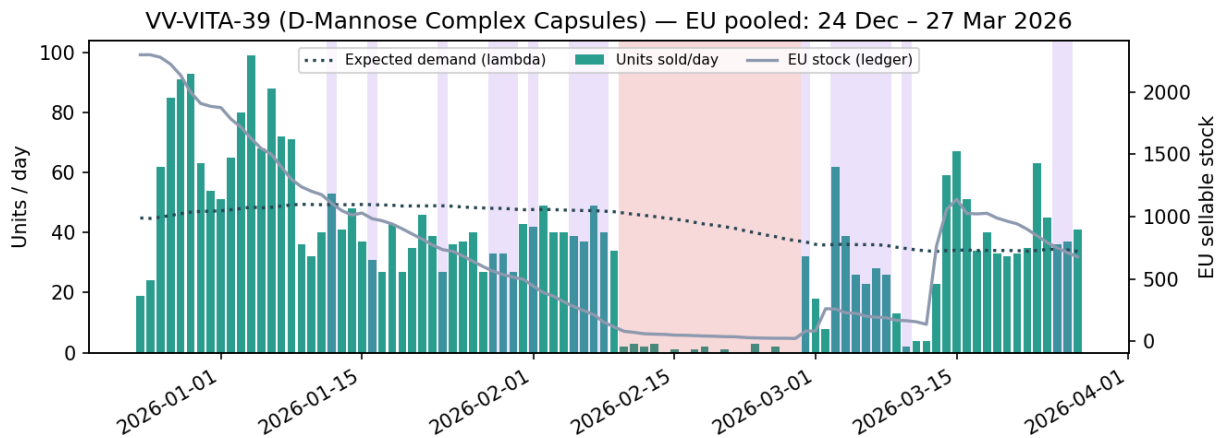
Lost / Miss revenue = missed units x avg price

Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).

'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-39 — D-Mannose Complex Capsules, EU-wide (Pan-EU pooled). Full year: **18 OOS days** (lost €14.285 revenue / €4.180 CM3) and **21 cooling-down days** (miss €3.887 revenue / €1.138 CM3).



Highlighted window 24 Dec 2025 – 27 Mar 2026: EU stock ranged from ~2,299 down to ~22 units and reach bottomed at ~1 day(s). In this window the model flagged **18 OOS day(s)** (red/orange shading) and **21 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out. Stock was replenished on **13 Mar** (EU balance jumps in the chart), after which sales recover and the flags clear — the table's last rows show those first restock days.

What "cooling down" means here. On the purple days the SKU kept selling (units > 0) but demand was deliberately damped to stretch the remaining stock to the next shipment. **Before cooling down** it sold at ~€20.38 with ~€0/day of ad spend; **during the throttle** the price was lifted to ~€21.05 and ad spend cut to ~€23/day. Sales fell to about 33/day versus the expected ~46/day, and that gap is booked as a **voluntary miss** — a pricing / advertising choice to protect availability and the listing's ranking — not a lost stock-out. Crucially the SKU keeps selling: the moment a throttle pushes sales to zero it is no longer cooling down and counts as OOS (lost). Cooling down trades a little volume now (the miss) to avoid the larger ranking and lost-sales hit of a full stock-out.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
01-12	53	49	1106	20	21.05	Cooling down		
01-16	31	49	984	18	21.04	Cooling down		105
01-23	27	49	735	13	21.06	Cooling down		125
01-28	33	48	562	11	21.26	Cooling down		86
01-29	33	48	535	11	21.18	Cooling down		86
02-10	2	47	80	2	18.99	Critically low (<3d)	255	
02-11	3	46	71	2	22.57	Critically low (<3d)	245	
02-12	2	46	60	2	20.95	Critically low (<3d)	248	
02-13	3	45	57	2	21.90	Critically low (<3d)	240	
02-14	0	45	55	2	—	Critically low (<3d)	254	
02-15	1	45	49	2	22.90	Critically low (<3d)	245	
03-12	4	34	136	17	18.02	—		
03-13	23	34	764	85	19.22	—		
03-14	59	34	1057	116	18.14	—		
03-15	67	34	1141	115	18.47	—		
03-16	51	34	1026	73	17.79	—		

6. Caveats & rules

Available stock = on-hand sellable + units in transit between FCs, so a Pan-EU redistribution (e.g. a bulk receipt landing at the Poland hub, then moving to other FCs over ~1-3 weeks) is not read as a stock-out. **Customer returns do not end a stock-out** — once OOS, a SKU stays OOS through return-driven balance blips until a genuine inbound Receipt or sales recover to $\geq$ half of expected. **GB** is a separate (non-Pan-EU) warehouse, analysed on its own.

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.